1 Gain law

(bars denote normalized quantities: lengths by R; $a_o := \Delta t/(\gamma_N R)$ [1/pN]; $\bar{a} := a/a_{\text{nom}}$; $\bar{f} := a_o f$, so that $\bar{a}\,\bar{f} = \Delta\bar{w}$; $\bar{w}$ = particle centre along the wall normal; $\bar{w}_s$ = the known reference surface on the same axis (plane: the wall, so $\bar{w} - \bar{w}_s = \bar{h}$ with contact at 1; cell: the visible surface); δ = depth of the effective (hydrodynamic) wall below $\bar{w}_s$, in R, the one constant of the wall; $g := (\bar{w} - \bar{w}_s) + \delta$ = centre distance to the effective wall, the law is used on $g \geq 1.1$; $u := 1/g$; $e_\theta = \theta - \hat{\theta}$)

$$D_\perp(u) = 1 - \tfrac{9}{8}u + \tfrac{1}{2}u^3 - \tfrac{57}{100}u^4 + \tfrac{1}{5}u^5 + \tfrac{7}{200}u^{11} - \tfrac{1}{25}u^{12}$$

$$\bar{a}(\bar{w};\,\delta) = D_\perp\Big(\frac{1}{(\bar{w} - \bar{w}_s) + \delta}\Big)$$

$$\bar{a}'(\bar{w};\,\delta) := \frac{\partial \bar{a}}{\partial \bar{w}} = -\,u^2\,D'_\perp(u)$$

$$\bar{a}''(\bar{w};\,\delta) := \frac{\partial^2 \bar{a}}{\partial \bar{w}^2} = u^4\,D''_\perp(u) + 2u^3\,D'_\perp(u)$$

$$\Big(\frac{\partial \bar{a}}{\partial \delta}\Big)_{\bar{w}} = \bar{a}', \qquad \Big(\frac{\partial \bar{a}}{\partial \bar{w}_s}\Big)_{\bar{w}} = -\,\bar{a}', \qquad \frac{\partial \bar{a}'}{\partial \delta} = \bar{a}''$$

$$D_\perp(1) = 0, \qquad -\,D'_\perp(1) = 1, \qquad \lim_{u\to 0}\frac{1 - D_\perp(u)}{u} = \tfrac{9}{8}$$

$$\bar{a}' = \Lambda(\bar{a}) := -\,u(\bar{a})^2\,D'_\perp\big(u(\bar{a})\big), \qquad u(\bar{a}) := D_\perp^{-1}(\bar{a}) \text{ on } u \in (0,1], \qquad \frac{\partial \Lambda}{\partial \delta} = 0$$

$$\frac{d\Lambda}{d\bar{a}} = \frac{\bar{a}''}{\bar{a}'}$$

$$a_w = a_o\,\bar{a}_w$$

2 True dynamics

$\big(\delta\bar{w}_j[k] := \delta\bar{w}[k - d + j - 1],\ j = 1, \ldots, d+1,\ d = 2\big)$

$$\bar{w}[k] = \bar{w}_d[k] - \delta\bar{w}_3[k]$$

$$\bar{w}_d[k+1] = \bar{w}_d[k] + \Delta\bar{w}_d[k]$$

$$\bar{a}_w[k] = D_\perp\Big(\frac{1}{(\bar{w}[k] - \bar{w}_s) + \delta}\Big)$$

$$\delta\bar{w}_1[k+1] = \delta\bar{w}_2[k]$$

$$\delta\bar{w}_2[k+1] = \delta\bar{w}_3[k]$$

$$\delta\bar{w}_3[k+1] = \lambda_c\,\delta\bar{w}_3[k] - F_{dw}[k]\,e_{\bar{a}_w}[k] - \varepsilon_w[k]$$

$$\Delta\bar{w}[k] = \Delta\bar{w}_d[k] + (1 - \lambda_c)\,\delta\bar{w}_3[k] + F_{dw}[k]\,e_{\bar{a}_w}[k] + \varepsilon_w[k]$$

$$\delta[k+1] = \delta[k]$$

$$F_{dw}[k] = \bar{f}_{dw}[k] + (1-\lambda_c)\sum_{i=1}^{d} \bar{f}_{dw}[k-i]$$

$$\varepsilon_w[k] = \bar{a}_w[k]\,\bar{f}_T[k] + (1-\lambda_c)\sum_{i=1}^{d} \bar{a}_w[k-i]\,\bar{f}_T[k-i] - (1-\lambda_c)\,n_w[k]$$

3 Estimator A: δ as a state, gain computed (no a slot)

$$\mathbf{x}_A = \big[\,\delta\bar{w}_1,\ \delta\bar{w}_2,\ \delta\bar{w}_3,\ \delta\,\big]^\top$$

$$\begin{aligned}
\hat{\bar{w}}[k] &= \bar{w}_d[k] - \delta\hat{\bar{w}}_3[k] \\
\hat{g}[k] &= (\hat{\bar{w}}[k] - \bar{w}_s) + \hat{\delta}[k] \\
\hat{\bar{a}}_w[k] &= D_\perp\Big(\frac{1}{\hat{g}[k]}\Big) \\
\delta\hat{\bar{w}}_1[k+1] &= \delta\hat{\bar{w}}_2[k] \\
\delta\hat{\bar{w}}_2[k+1] &= \delta\hat{\bar{w}}_3[k] \\
\delta\hat{\bar{w}}_3[k+1] &= \lambda_c\,\delta\hat{\bar{w}}_3[k] \\
\hat{\delta}[k+1] &= \hat{\delta}[k]
\end{aligned}$$

4 Error dynamics A

$\big(\mathbf{e}_A = [\,e_{\delta\bar{w}_1},\ e_{\delta\bar{w}_2},\ e_{\delta\bar{w}_3},\ e_\delta\,]^\top$; coefficients at the estimate, $\bar{a}'[k] := \bar{a}'(\hat{g}[k])\big)$

$$\begin{aligned}
e_{\bar{a}_w}[k] &= \bar{a}'[k]\,\big(e_\delta[k] - e_{\delta\bar{w}_3}[k]\big) \\
e_{\delta\bar{w}_1}[k+1] &= e_{\delta\bar{w}_2}[k] \\
e_{\delta\bar{w}_2}[k+1] &= e_{\delta\bar{w}_3}[k] \\
e_{\delta\bar{w}_3}[k+1] &= \big(\lambda_c + F_{dw}[k]\,\bar{a}'[k]\big)\,e_{\delta\bar{w}_3}[k] - F_{dw}[k]\,\bar{a}'[k]\,e_\delta[k] - \varepsilon_w[k] \\
e_\delta[k+1] &= e_\delta[k]
\end{aligned}$$

$$F_e^A[k] = \begin{bmatrix}
0 & 1 & 0 & 0 \\
0 & 0 & 1 & 0 \\
0 & 0 & \lambda_c + F_{dw}[k]\,\bar{a}'[k] & -F_{dw}[k]\,\bar{a}'[k] \\
0 & 0 & 0 & 1
\end{bmatrix}$$

$$\mathrm{Var}\big(e_{\bar{a}_w}[k]\big) = \bar{a}'[k]^2\,\big(P_{33}[k] + P_{44}[k] - 2P_{34}[k]\big)$$

5 Estimator B: a slot carried along the autonomous law (no δ state)

$\big(M[k] := \Delta\bar{w}_d[k] + (1-\lambda_c)\,\delta\hat{\bar{w}}_3[k]$; Λ does not contain δ, so δ has no column in any row of B and enters only through $\hat{\bar{a}}_w[0]$ and $P_{44}[0]\big)$

$$\mathbf{x}_B = \big[\,\delta\bar{w}_1,\ \delta\bar{w}_2,\ \delta\bar{w}_3,\ \bar{a}_w\,\big]^\top$$

$$\begin{aligned}
\delta\hat{\bar{w}}_1[k+1] &= \delta\hat{\bar{w}}_2[k] \\
\delta\hat{\bar{w}}_2[k+1] &= \delta\hat{\bar{w}}_3[k] \\
\delta\hat{\bar{w}}_3[k+1] &= \lambda_c\,\delta\hat{\bar{w}}_3[k] \\
\hat{\bar{a}}_w[k+1] &= D_\perp\Big(\frac{1}{1/u(\hat{\bar{a}}_w[k]) + M[k]}\Big) \\
&= \hat{\bar{a}}_w[k] + \Lambda\big(\hat{\bar{a}}_w[k]\big)\,M[k] + \tfrac{1}{2}\,\bar{a}''\big(\hat{\bar{a}}_w[k]\big)\,M[k]^2 + \dots
\end{aligned}$$

6 Error dynamics B

$\big(\mathbf{e}_B = [\,e_{\delta\bar{w}_1},\ e_{\delta\bar{w}_2},\ e_{\delta\bar{w}_3},\ e_{\bar{a}_w}\,]^\top$; first-order step, coefficients at the estimate$\big)$

$$\begin{aligned}
e_{\delta\bar{w}_1}[k+1] &= e_{\delta\bar{w}_2}[k] \\
e_{\delta\bar{w}_2}[k+1] &= e_{\delta\bar{w}_3}[k] \\
e_{\delta\bar{w}_3}[k+1] &= \lambda_c\,e_{\delta\bar{w}_3}[k] - F_{dw}[k]\,e_{\bar{a}_w}[k] - \varepsilon_w[k] \\
e_{\bar{a}_w}[k+1] &= (1-\lambda_c)\,\Lambda\big(\hat{\bar{a}}_w[k]\big)\,e_{\delta\bar{w}_3}[k] + \Big(1 + \Lambda\big(\hat{\bar{a}}_w[k]\big)\,F_{dw}[k] + \frac{\bar{a}''(\hat{\bar{a}}_w[k])}{\bar{a}'(\hat{\bar{a}}_w[k])}\,M[k]\Big)e_{\bar{a}_w}[k] \\
&\quad + \Lambda\big(\hat{\bar{a}}_w[k]\big)\,\varepsilon_w[k]
\end{aligned}$$

$$F_e^B[k] = \begin{bmatrix}
0 & 1 & 0 & 0 \\
0 & 0 & 1 & 0 \\
0 & 0 & \lambda_c & -F_{dw}[k] \\
0 & 0 & (1-\lambda_c)\,\Lambda(\hat{\bar{a}}_w[k]) & 1 + \Lambda(\hat{\bar{a}}_w[k])\,F_{dw}[k] + \dfrac{\bar{a}''(\hat{\bar{a}}_w[k])}{\bar{a}'(\hat{\bar{a}}_w[k])}\,M[k]
\end{bmatrix}$$

7 Measurement

$\big(\nabla_d\bar{w}_d[k] := \bar{w}_d[k] - \bar{w}_d[k-d]$; $\hat{\bar{w}}_d[k] := \hat{\bar{w}}[k] - \nabla_d\bar{w}_d[k]$; $\hat{g}_d[k] := (\hat{\bar{w}}_d[k] - \bar{w}_s) + \hat{\delta}[k]\big)$

$$\begin{aligned}
y_1[k] &= \delta\bar{w}_1[k] + n_w[k] \\
y_2[k] &= \bar{a}_w[k-d] + n_a[k] \\
\hat{y}_1[k] &= \delta\hat{\bar{w}}_1[k]
\end{aligned}$$

A: $\hat{\bar{a}}_d[k] = D_\perp\Big(\dfrac{1}{\hat{g}_d[k]}\Big)$

$\hat{y}_2[k] = \hat{\bar{a}}_d[k]$

$H^A[k] = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & -\bar{a}'(\hat{g}_d[k]) & \bar{a}'(\hat{g}_d[k]) \end{bmatrix}$

B: $\hat{\bar{a}}_d[k] = D_\perp\Big(\dfrac{1}{1/u(\hat{\bar{a}}_w[k]) - \nabla_d \bar{w}_d[k]}\Big)$

$\hat{y}_2[k] = \hat{\bar{a}}_d[k]$

$\dfrac{\partial \hat{\bar{a}}_d[k]}{\partial \hat{\bar{a}}_w[k]} = \dfrac{\Lambda(\hat{\bar{a}}_d[k])}{\Lambda(\hat{\bar{a}}_w[k])}$

$H^B[k] = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & \dfrac{\Lambda(\hat{\bar{a}}_d[k])}{\Lambda(\hat{\bar{a}}_w[k])} \end{bmatrix}$

$\mathbf{e}_y[k] = H[k]\,\mathbf{e}^-[k] + \begin{bmatrix} n_w[k] \\ n_a[k] \end{bmatrix}$

8 Noise

$Q^A[k] = Q_{33}[k]\,\mathbf{g}_A\,\mathbf{g}_A^\top, \qquad \mathbf{g}_A = \begin{bmatrix} 0,\ 0,\ -1,\ 0 \end{bmatrix}^\top$

$Q^B[k] = Q_{33}[k]\,\mathbf{g}_B\,\mathbf{g}_B^\top, \qquad \mathbf{g}_B = \begin{bmatrix} 0,\ 0,\ -1,\ \Lambda(\hat{\bar{a}}_w[k]) \end{bmatrix}^\top$

$Q_{33}[k] = \kappa_T\Big(\hat{\bar{a}}_w[k] + (1-\lambda_c)^2\big(\hat{\bar{a}}_w[k-1] + \hat{\bar{a}}_w[k-2]\big)\Big) + (1-\lambda_c)^2\sigma_{n_w}^2, \qquad \kappa_T = \dfrac{4k_B T\,a_o}{R}$

$R[k] = \mathrm{diag}\big(R_1,\ R_2[k]\big), \qquad R_1 = \dfrac{\sigma_n^2}{R^2}, \qquad R_2[k] = 2a_{\mathrm{cov}}^2\,IF_{\mathrm{var}}\big(\hat{\bar{a}}_w[k]\big)\big(\hat{\bar{a}}_w[k] + \xi\big)^2$

9 Observability

($\Phi(k,k_0) := F_e[k-1]\cdots F_e[k_0]$, $\Phi(k_0,k_0)=I$; N window length)

($O_w := R_w^{-1/2}\,[\,H[k_0]\Phi(k_0,k_0);\ \ldots;\ H[k_0+N]\Phi(k_0+N,k_0)\,]$; $O_w\,\mathbf{v}_i=\sigma_i\,\mathbf{u}_i$; $\sigma_\theta := \sqrt{[(O_w^\top O_w)^{-1}]_{\theta\theta}}$)

Gate 1, the two columns of e_δ (A):

$$F^A_{e,34}[k] = -\,F_{dw}[k]\,\bar a'(\hat g[k]), \qquad H^A_{24}[k] = \bar a'(\hat g_d[k])$$

Gate 2, time signatures of the pair ($e_{\delta\bar w_3},\ e_\delta$) (A):

$$e_\delta[k+1] = e_\delta[k]$$
$$e_{\delta\bar w_3}[k+1] = \big(\lambda_c + F_{dw}[k]\,\bar a'[k]\big)\,e_{\delta\bar w_3}[k] - F_{dw}[k]\,\bar a'[k]\,e_\delta[k] - \varepsilon_w[k], \qquad y_1[k+2] = e_{\delta\bar w_3}[k] + \ldots$$

Gate 3, $\det O_2$ of the pair with coefficients frozen ($\alpha := \lambda_c + F_{dw}\,\bar a'(\hat g)$, $\beta := F_{dw}\,\bar a'(\hat g)$, $p := \bar a'(\hat g_d)$):

$$A_2 = \begin{bmatrix} \alpha & -\beta \\ 0 & 1 \end{bmatrix}$$

row y_1 only: $C_2 = \begin{bmatrix} 1,\ 0 \end{bmatrix}$, $\det \begin{bmatrix} C_2 \\ C_2A_2 \end{bmatrix} = -\,\beta = -\,F_{dw}\,\bar a'(\hat g)$

row y_2 only: $C_2 = \begin{bmatrix} -p,\ p \end{bmatrix}$, $\det \begin{bmatrix} C_2 \\ C_2A_2 \end{bmatrix} = p^2\,(\alpha-\beta-1) = -\,(1-\lambda_c)\,\bar a'(\hat g_d)^2$

$\det O_2 = 0$ (y_1) $\iff$ $F_{dw}=0$ (hold) or $\bar a'(\hat g)=0$ (far field)
$\det O_2 = 0$ (y_2) $\iff$ $\bar a'(\hat g_d)=0$ (far field only)

Aliases (A):

$\Big(\dfrac{\partial\bar a}{\partial\delta}\Big)_{\bar w} + \Big(\dfrac{\partial\bar a}{\partial\bar w_s}\Big)_{\bar w} = 0$ $\implies$ only $\bar w_s - \delta$ enters; $\bar w_s$ is given, not a state
$\bar a'(\hat g)\to 0$ (far field) $\implies$ $F^A_{e,34}\to 0,\ H^A_{24}\to 0,\ e_\delta$ unobservable

B:

$$\frac{\partial\Lambda}{\partial\delta}=0,\quad \frac{\partial H^B}{\partial\delta}=0,\quad \frac{\partial F^B_e}{\partial\delta}=0$$
$\implies$ δ has no column in any row of B
$\implies$ gate 1 fails by construction: δ is not a state of B

Seed and prior

plant offsets: $\delta_{\text{plane}} = 0, \qquad \delta_{\text{cell}} = 1.05$ (`cell_cperp_delta`)

wall family prior (half the plane–cell spread, two-sided): $\sigma_{\delta,0} = 0.5$

arm T: $\hat{\delta}[0] = \delta_{\text{plant}}$

arm T $\pm$: $\hat{\delta}[0] = \delta_{\text{plant}} \pm 0.5$

arm W: $\bar{w}_s \to \bar{w}_s + 1, \qquad \hat{\delta}[0] = \delta_{\text{plant}}$

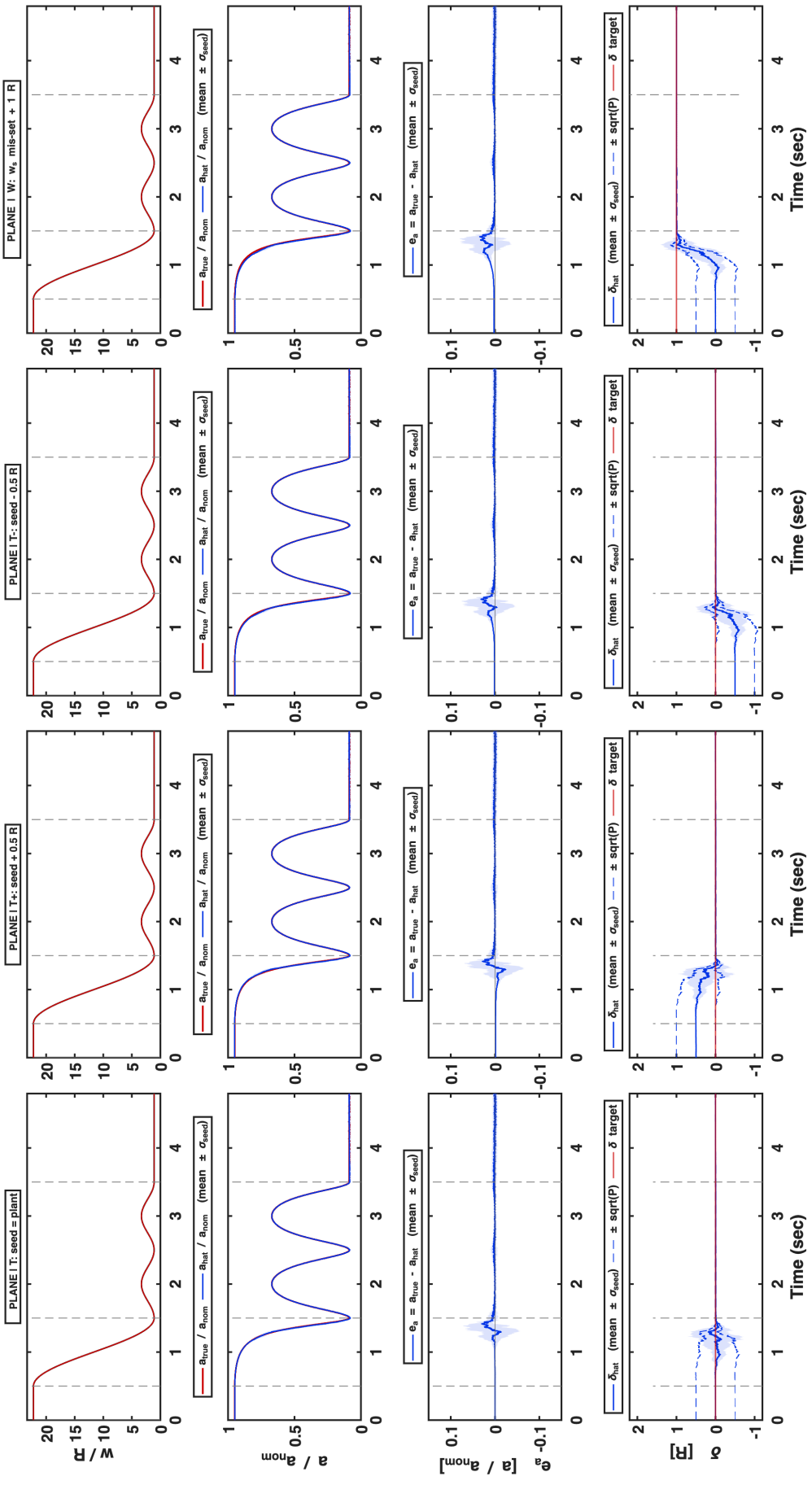
$$\hat{\bar{a}}_w[0] = D_{\perp}\Big(\frac{1}{(\bar{w}[0] - \bar{w}_s) + \hat{\delta}[0]}\Big)$$

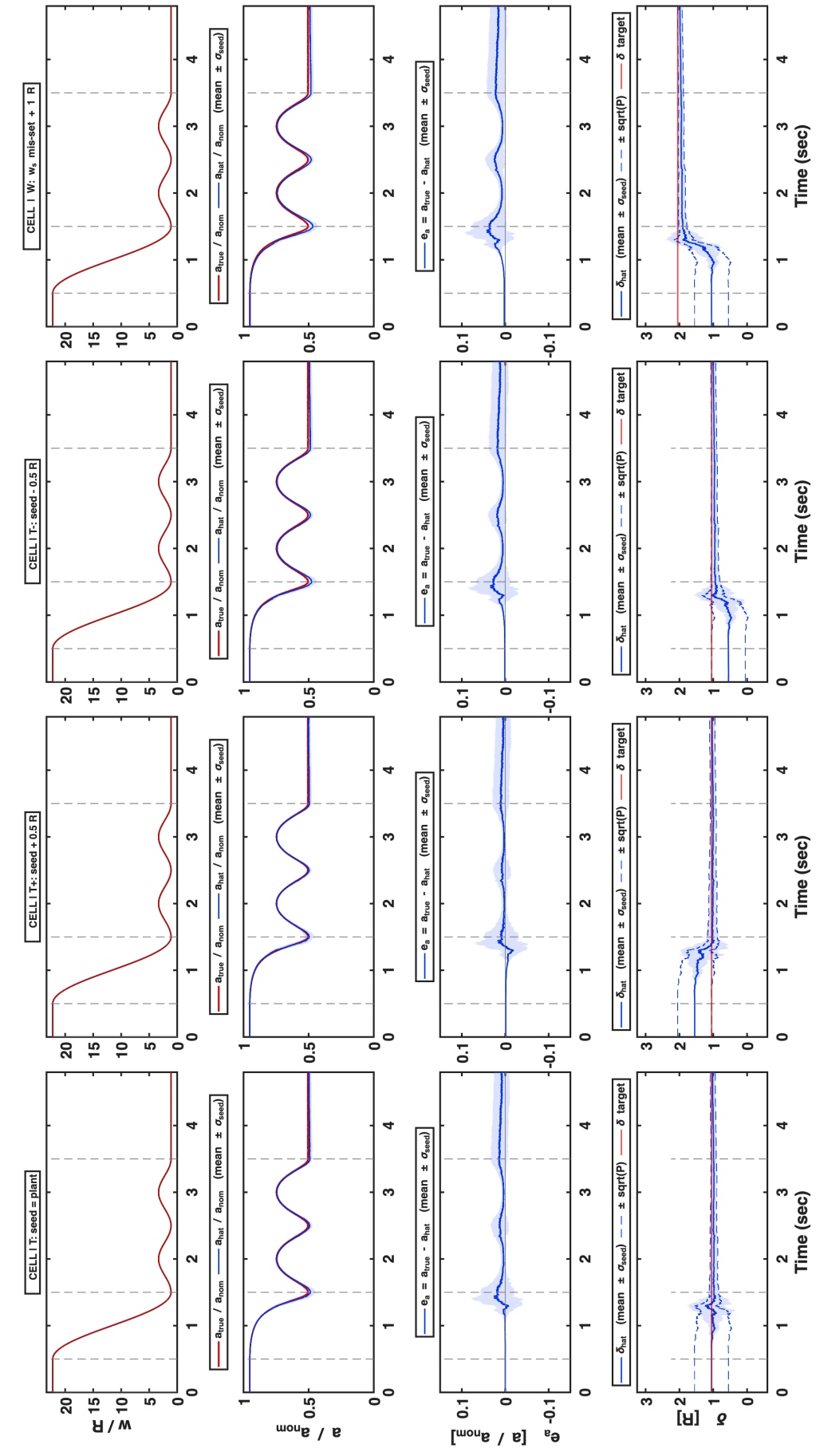
$$P[0] = \operatorname{diag}\big(P_{11}[0],\ P_{22}[0],\ P_{33}[0],\ P_{44}[0]\big)$$

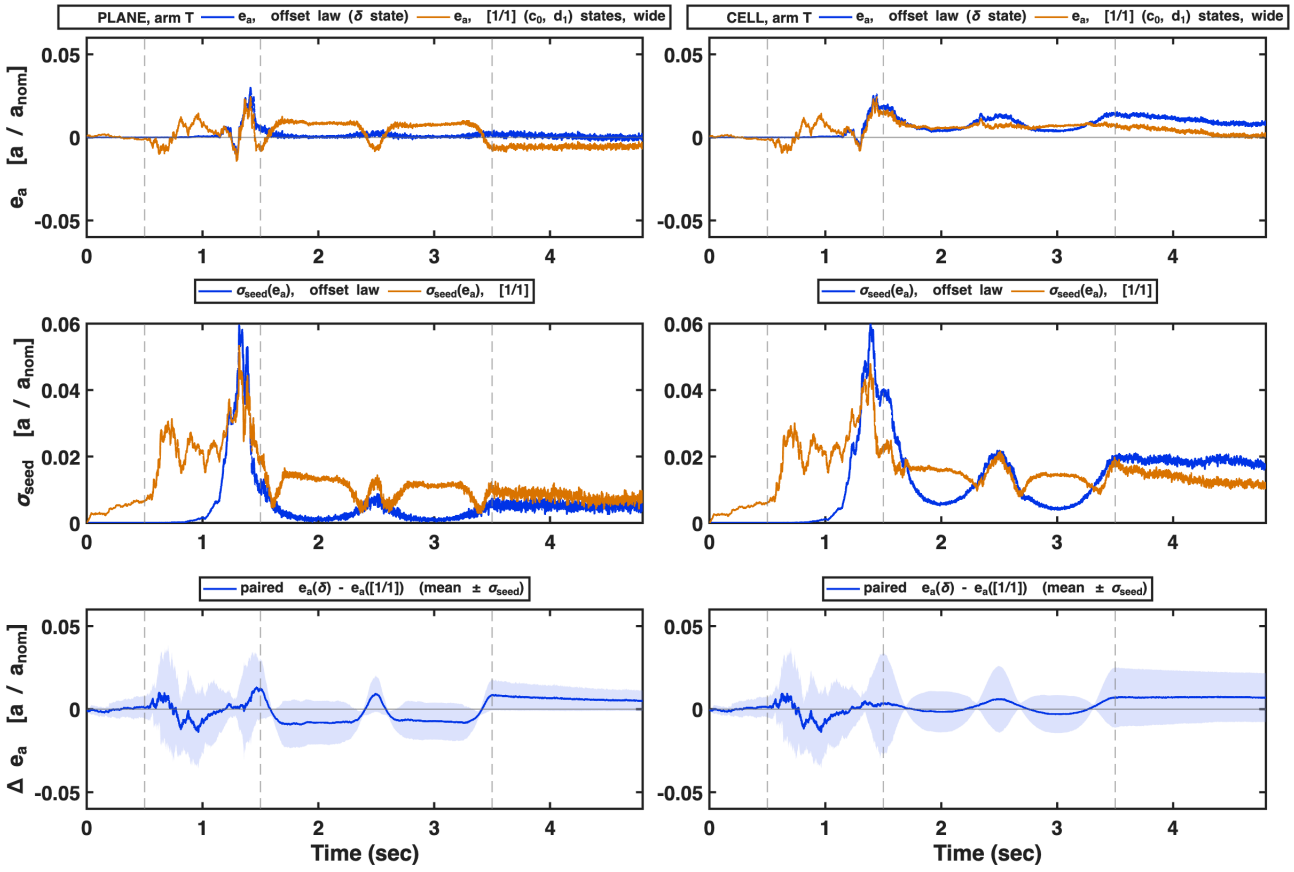
A: $\sqrt{P_{44}[0]} = \sigma_{\delta,0}, \qquad Q_{44} = 0$

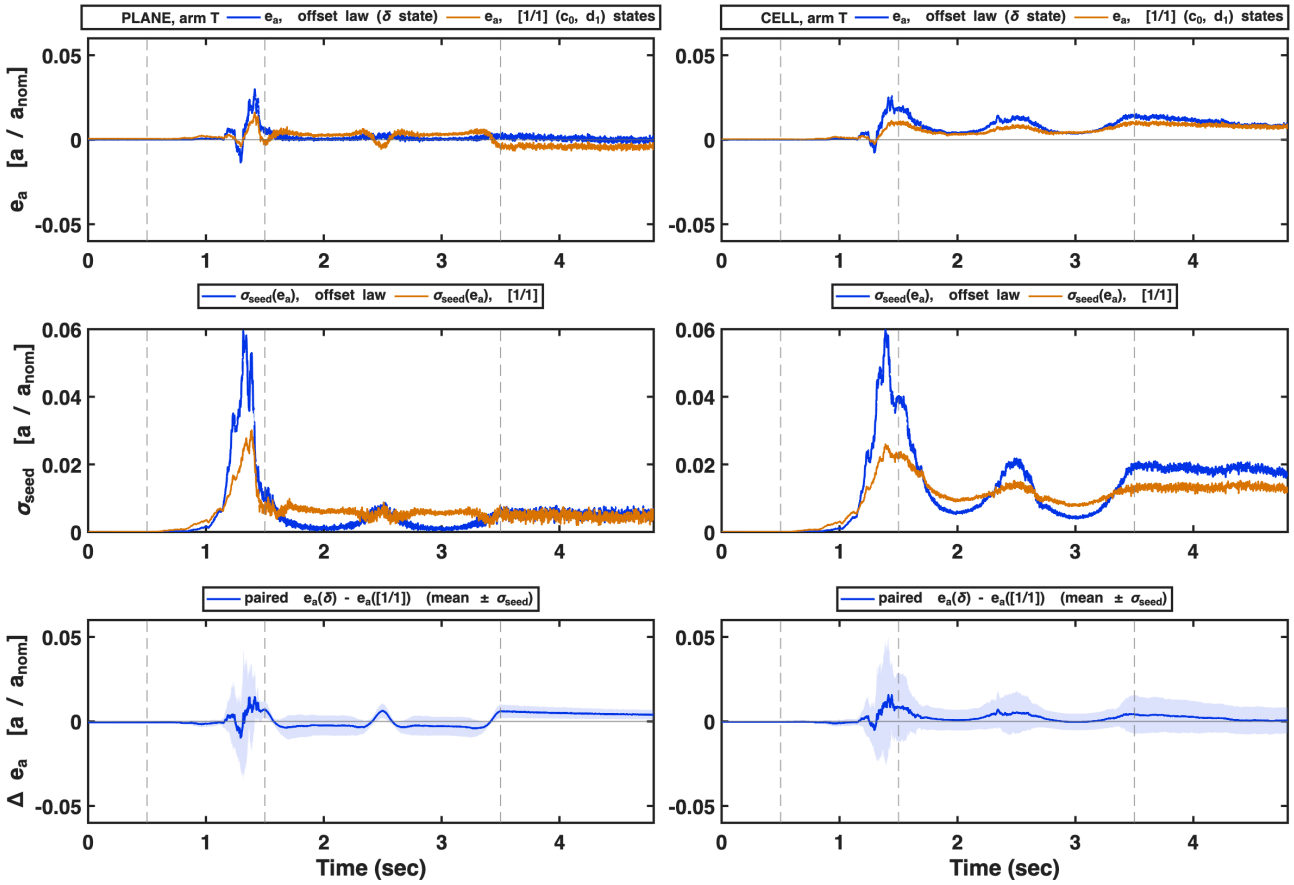
B: $P_{44}[0] = \bar{a}'\big(\hat{g}[0]\big)^2\, \sigma_{\delta,0}^2$

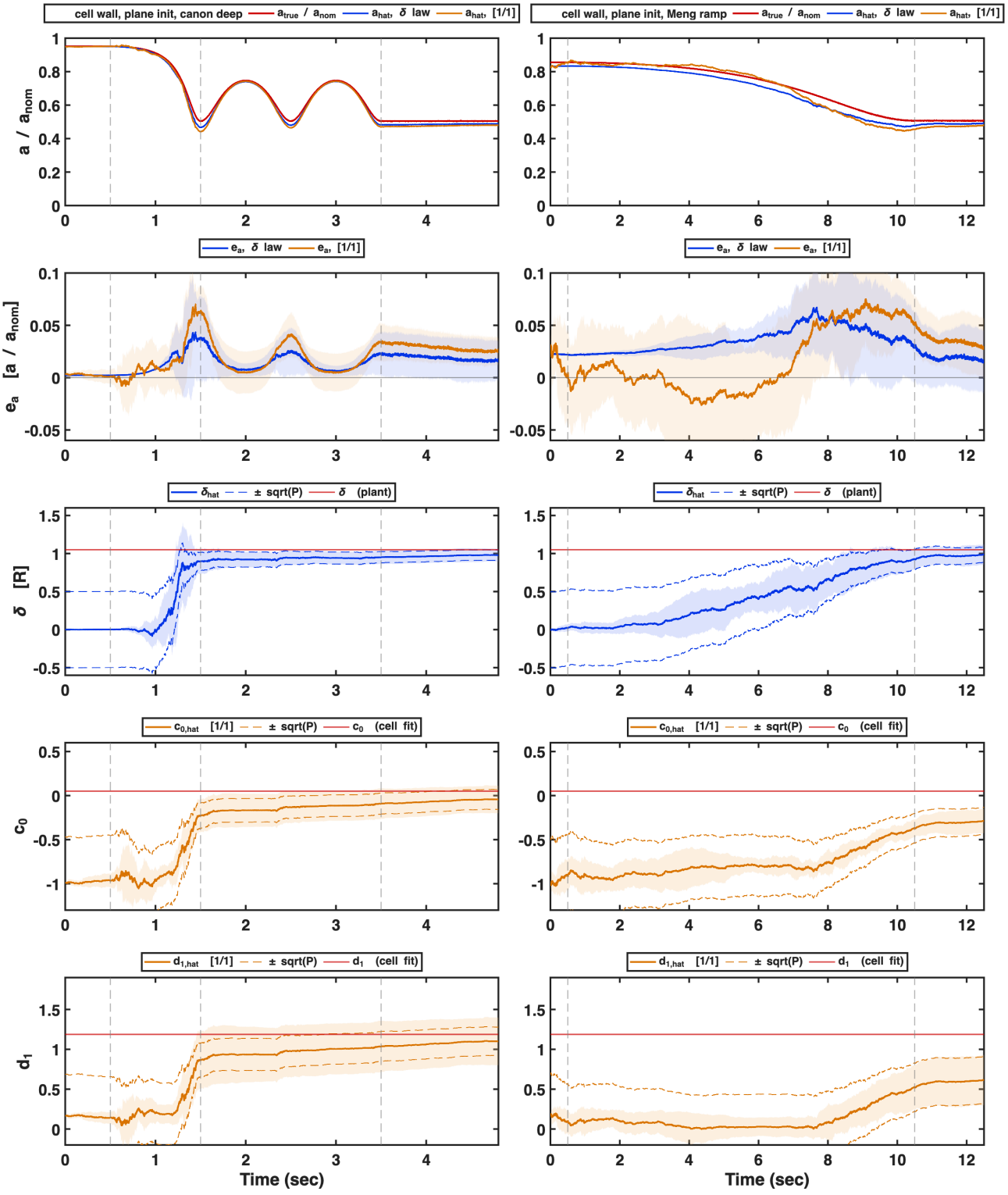
guard: $\hat{g}[k] \geq 1$ (the polynomial is evaluated at $\max(\hat{g}[k],\ 1)$), band $\hat{g} \geq 1.1$

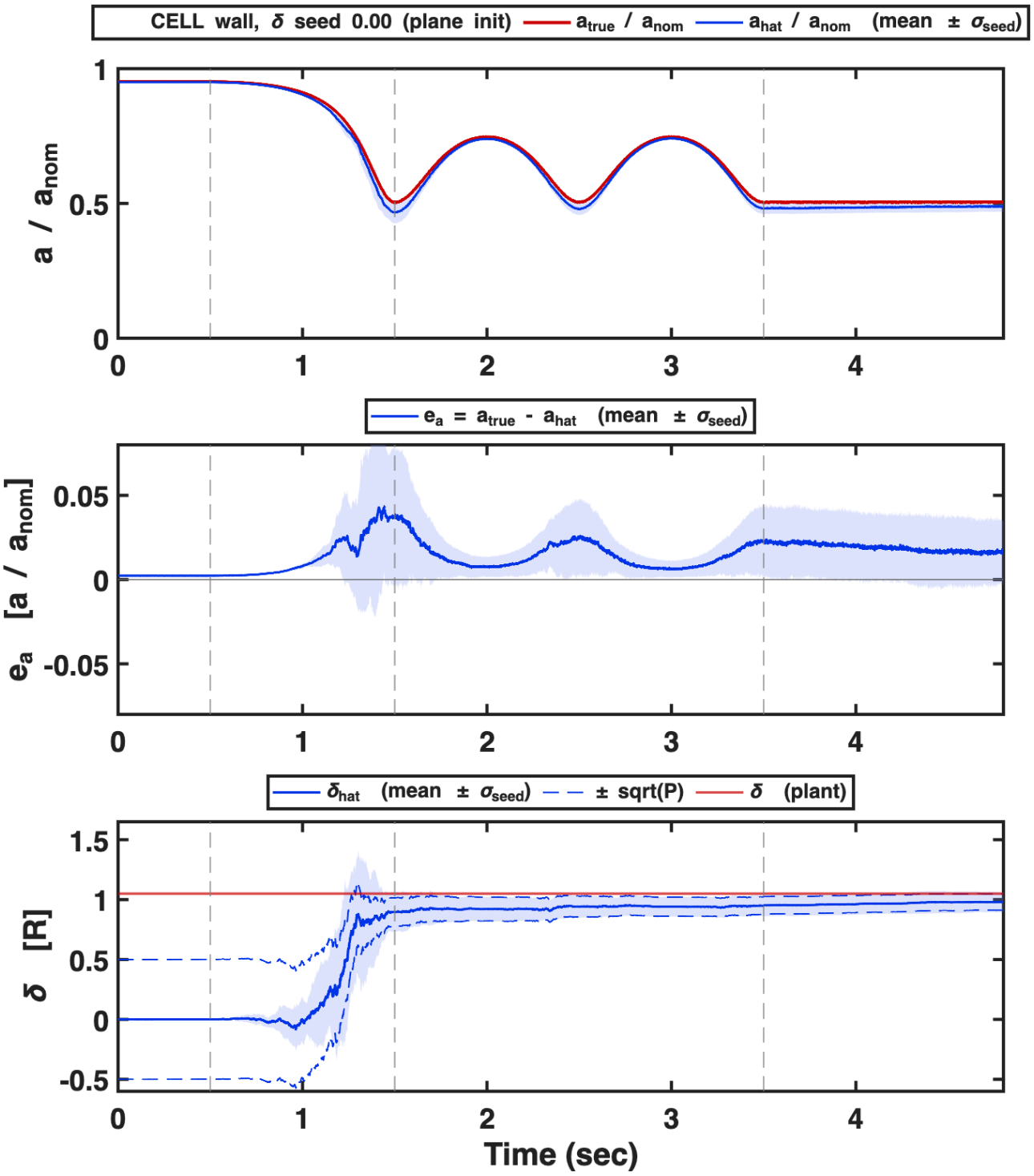

PLANE I T: seed = plant
PLANE I T+: seed + 0.5 R
PLANE I T-: seed - 0.5 R
PLANE I W: w_s mis-set + 1 R
W / R
a / a_nom
e_a [a / a_nom]
δ [R]
a_true / a_nom
a_hat / a_nom (mean ± σ_seed)
e_a = a_true - a_hat (mean ± σ_seed)
δ_hat (mean ± σ_seed)
± sqrt(P)
δ target
Time (sec)

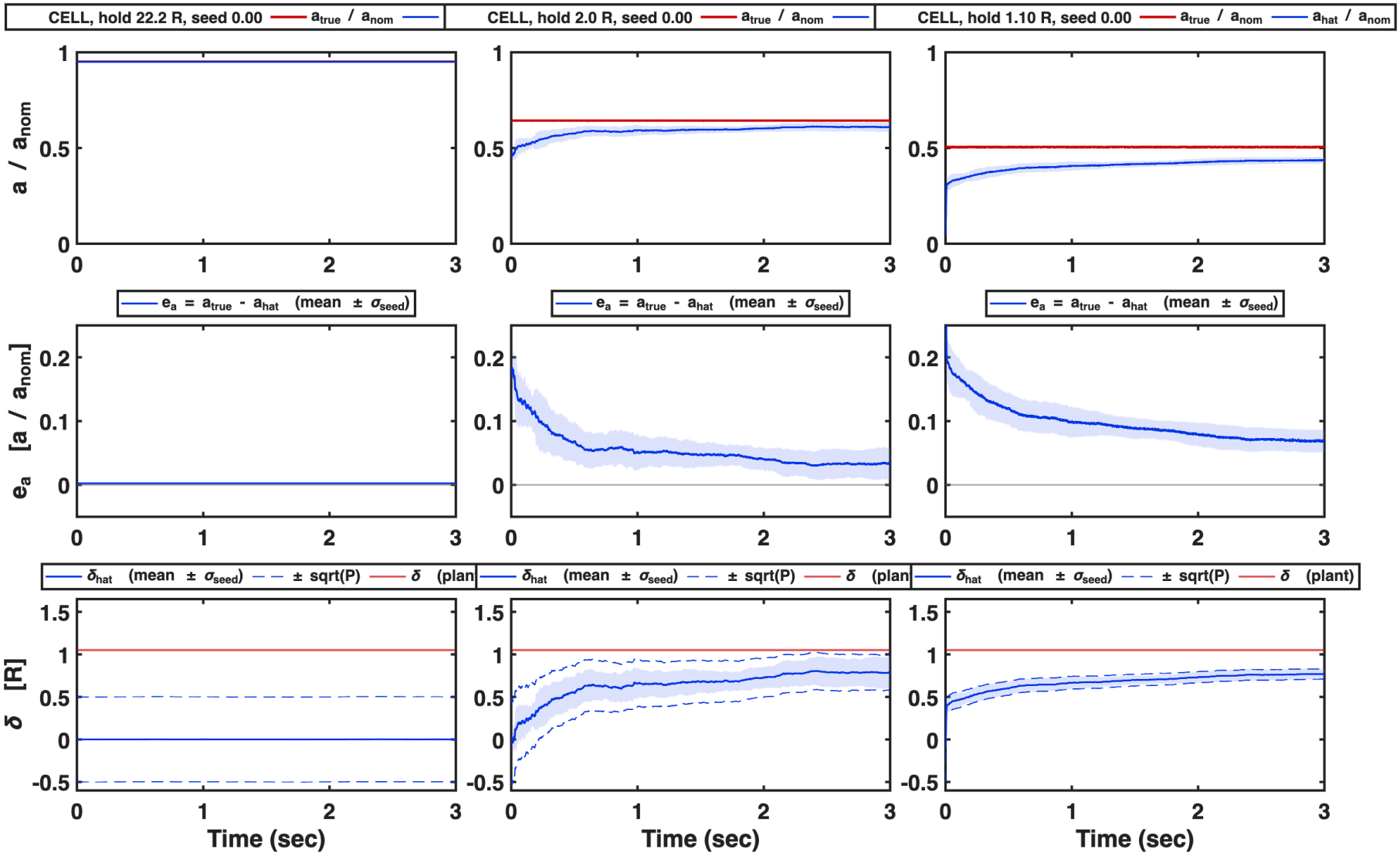

CELL | T: seed = plant
CELL | T+: seed + 0.5 R
CELL | T-: seed - 0.5 R
CELL | W: w_s mis-set + 1 R
W / R
a / a_{nom}
e_a [a / a_{nom}]
δ [R]
a_{true} / a_{nom}
a_{hat} / a_{nom} (mean ± σ_{seed})
e_a = a_{true} - a_{hat} (mean ± σ_{seed})
δ_{hat} (mean ± σ_{seed})
± sqrt(P)
δ target
Time (sec)

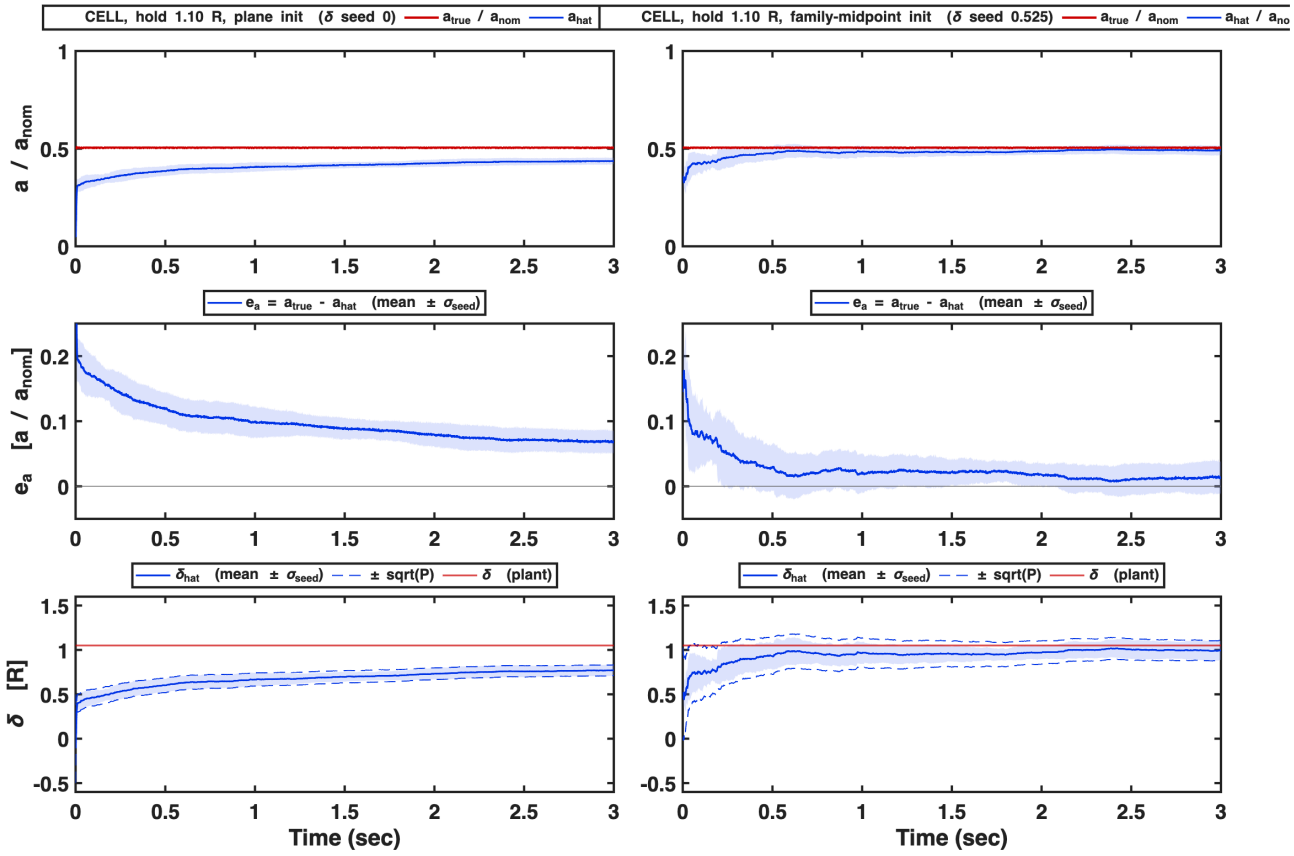

PLANE, arm T
CELL, arm T
Time (sec)
Time (sec)

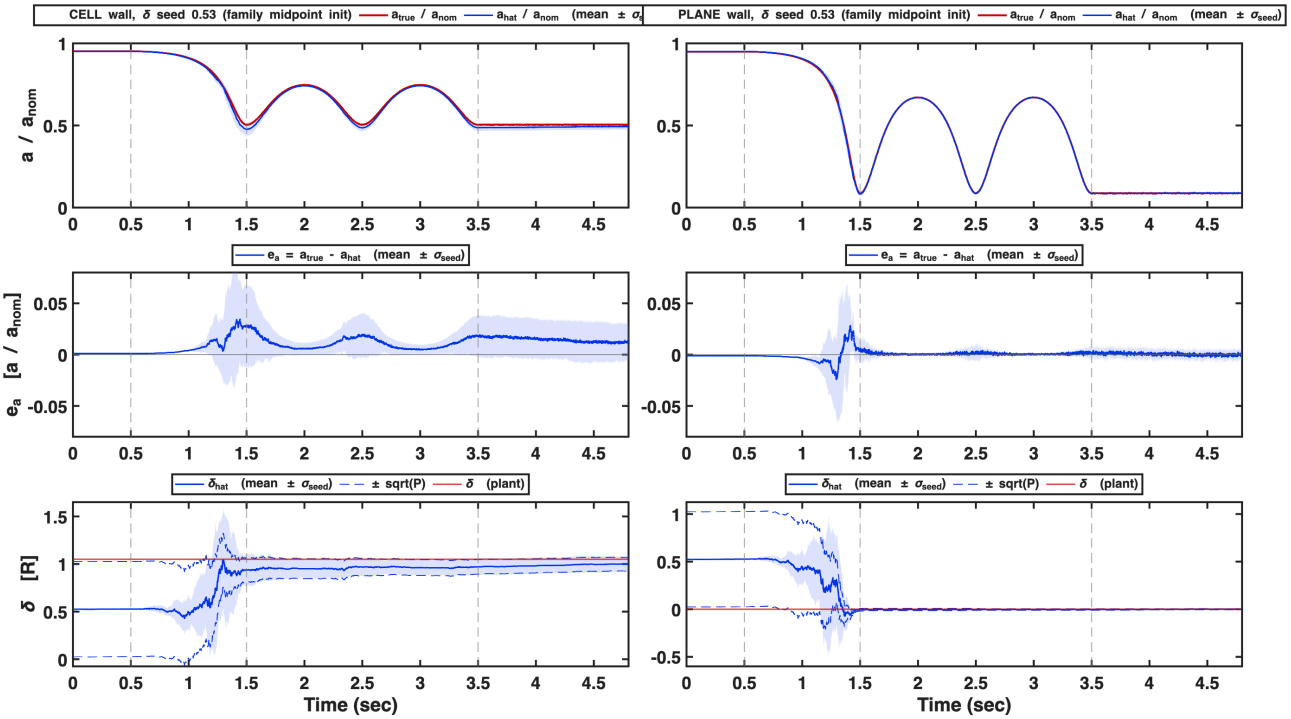

PLANE, arm T
CELL, arm T
Time (sec)
Time (sec)

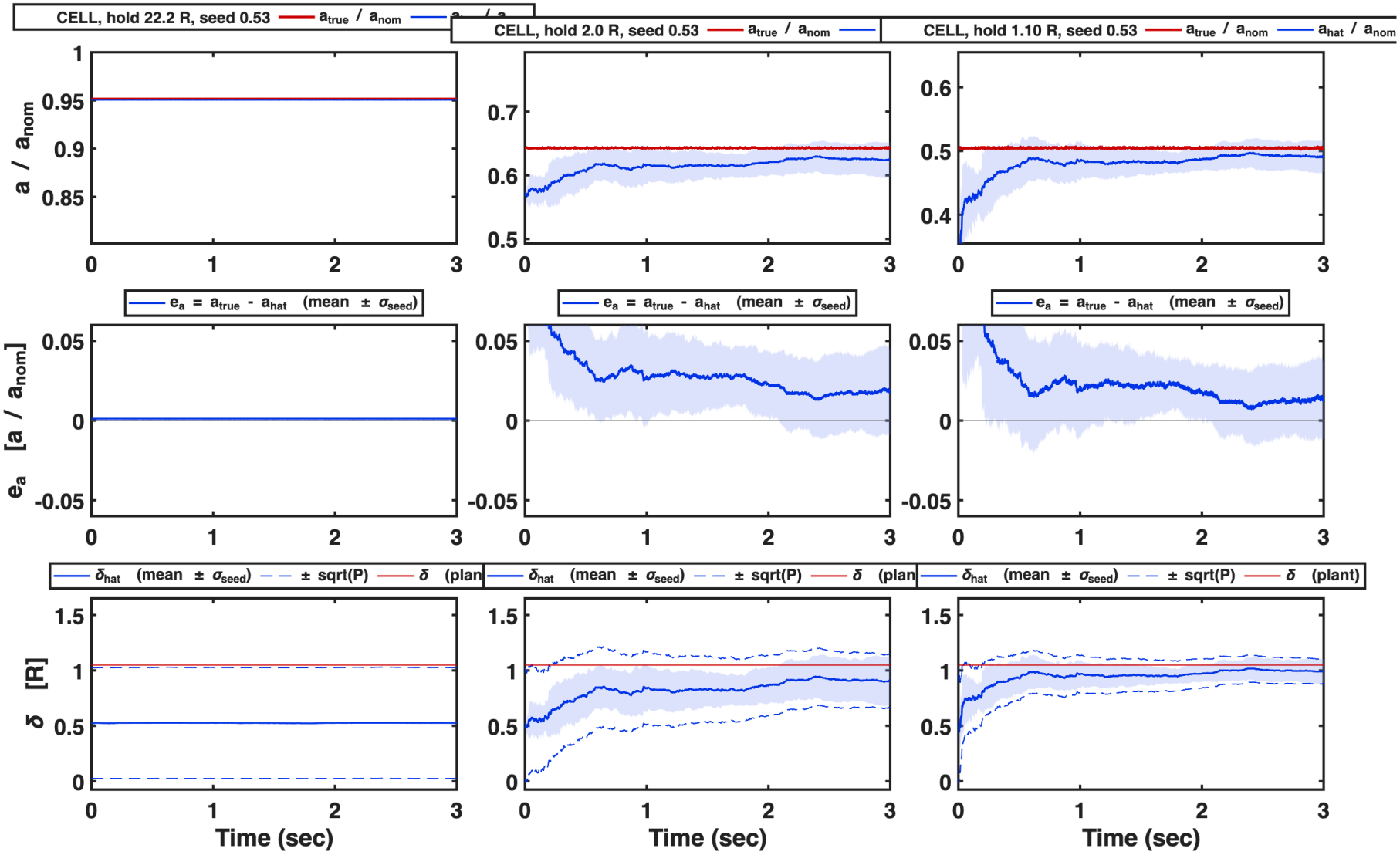

cell wall, plane init, canon deep
a_true / a_nom
a_hat, δ law
a_hat, [1/1]
cell wall, plane init, Meng ramp
a / a_nom
e_a, δ law
e_a, [1/1]
e_a [a / a_nom]
δ_hat ± sqrt(P) δ (plant)
δ [R]
c_0,hat [1/1] ± sqrt(P) c_0 (cell fit)
c_0
d_1,hat [1/1] ± sqrt(P) d_1 (cell fit)
d_1
Time (sec)

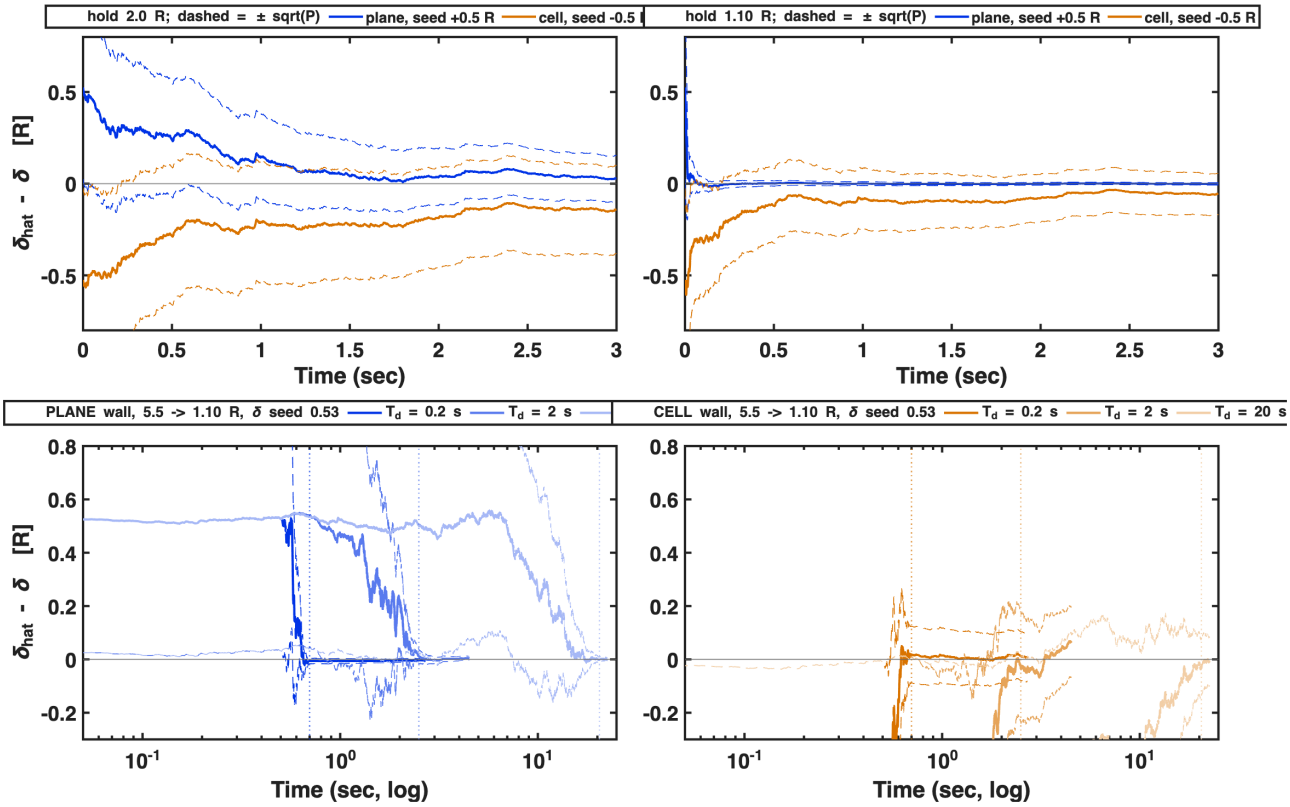

CELL wall, δ seed 0.00 (plane init) a_true / a_nom a_hat / a_nom (mean ± σ_seed)
a / a_nom
e_a = a_true - a_hat (mean ± σ_seed)
e_a [a / a_nom]
δ_hat (mean ± σ_seed) ± sqrt(P) δ (plant)
δ [R]
Time (sec)

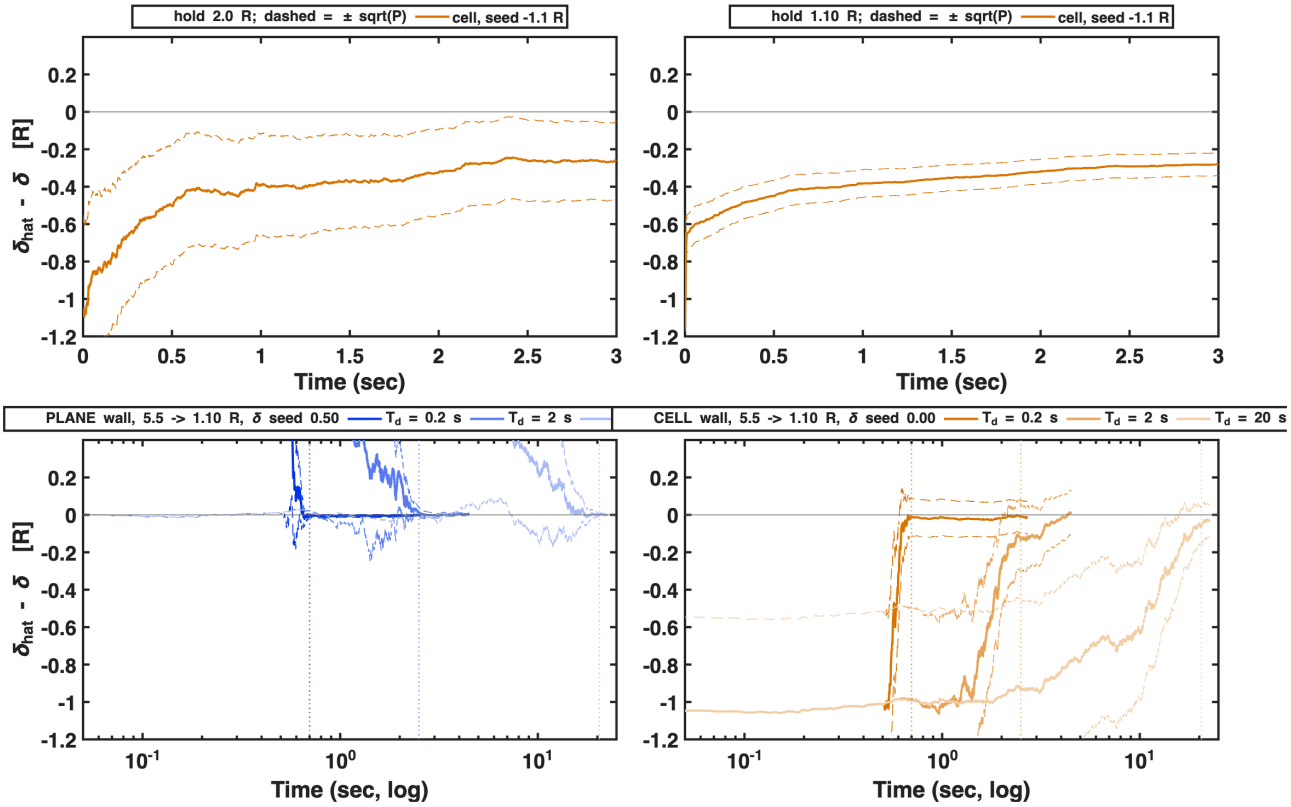

CELL, hold 22.2 R, seed 0.00
a_true / a_nom
CELL, hold 2.0 R, seed 0.00
a_true / a_nom
CELL, hold 1.10 R, seed 0.00
a_true / a_nom
a_hat / a_nom
a / a_nom
e_a = a_true - a_hat (mean ± σ_seed)
e_a [a / a_nom]
δ_hat (mean ± σ_seed)
± sqrt(P)
δ (plant)
δ [R]
Time (sec)

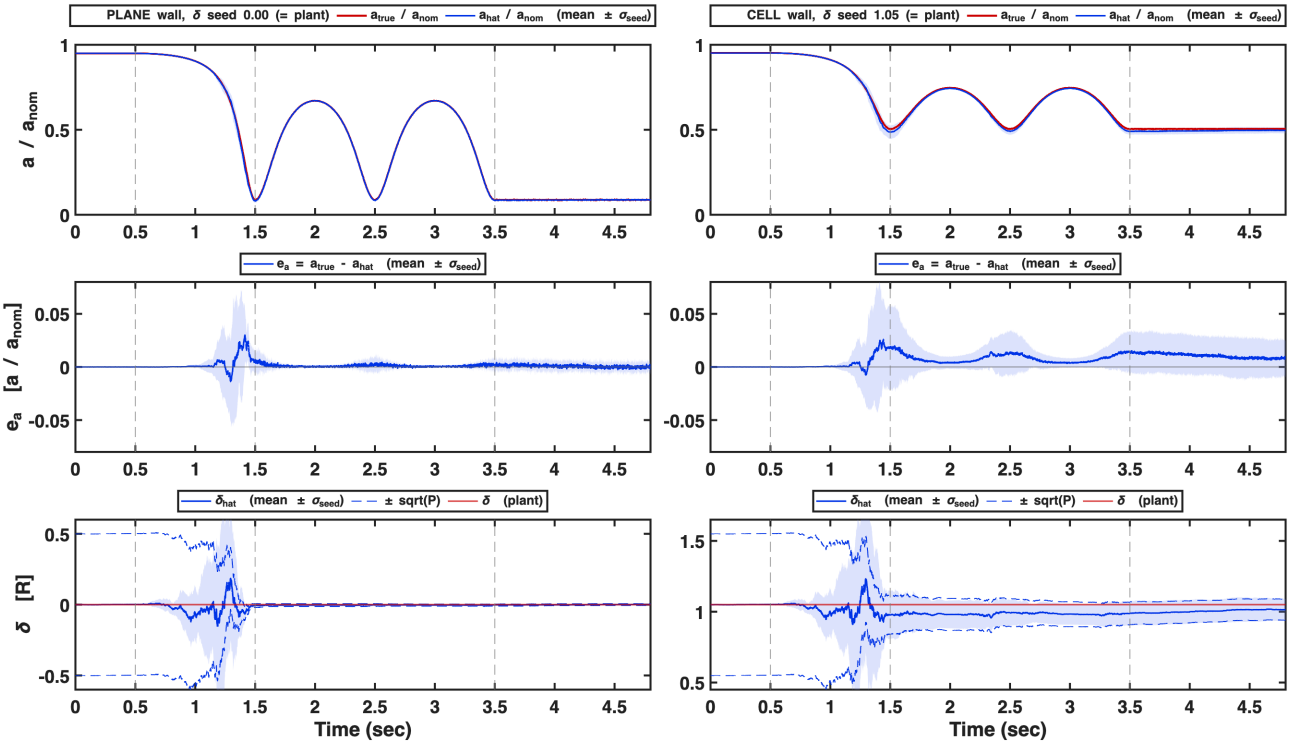

CELL, hold 1.10 R, plane init (δ seed 0) — a_true / a_nom — a_hat
CELL, hold 1.10 R, family-midpoint init (δ seed 0.525) — a_true / a_nom — a_hat / a_no
a / a_nom
e_a = a_true - a_hat (mean ± σ_seed)
e_a [a / a_nom]
δ_hat (mean ± σ_seed) ± sqrt(P) δ (plant)
δ [R]
Time (sec)
Time (sec)

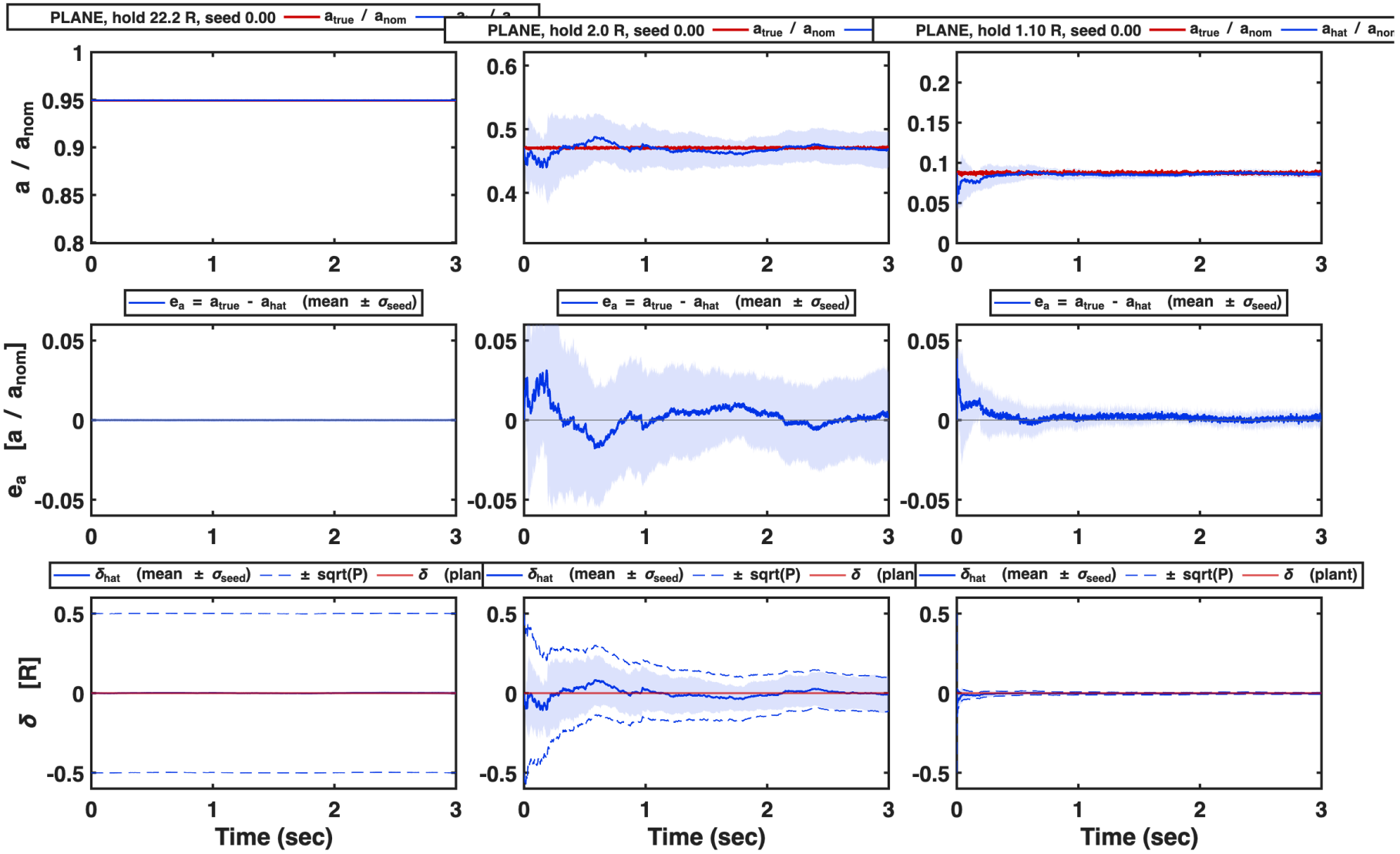

CELL wall, δ seed 0.53 (family midpoint init)
PLANE wall, δ seed 0.53 (family midpoint init)
a / a_nom
e_a [a / a_nom]
δ [R]
Time (sec)

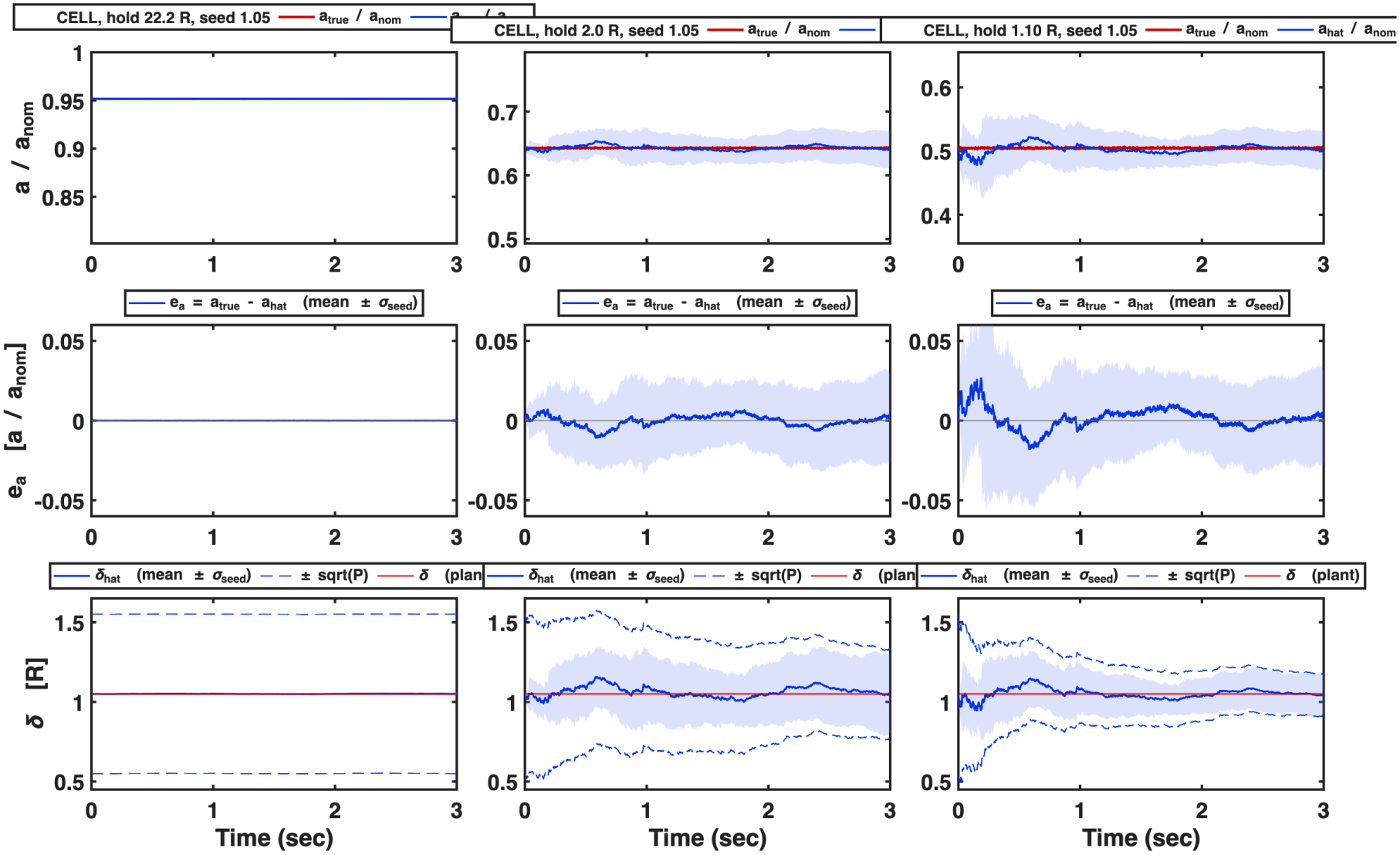

CELL, hold 22.2 R, seed 0.53
a_true / a_nom
CELL, hold 2.0 R, seed 0.53
a_true / a_nom
CELL, hold 1.10 R, seed 0.53
a_true / a_nom
a_hat / a_nom
a / a_nom
e_a = a_true - a_hat (mean ± σ_seed)
e_a [a / a_nom]
δ_hat (mean ± σ_seed)
± sqrt(P)
δ (plant)
δ [R]
Time (sec)

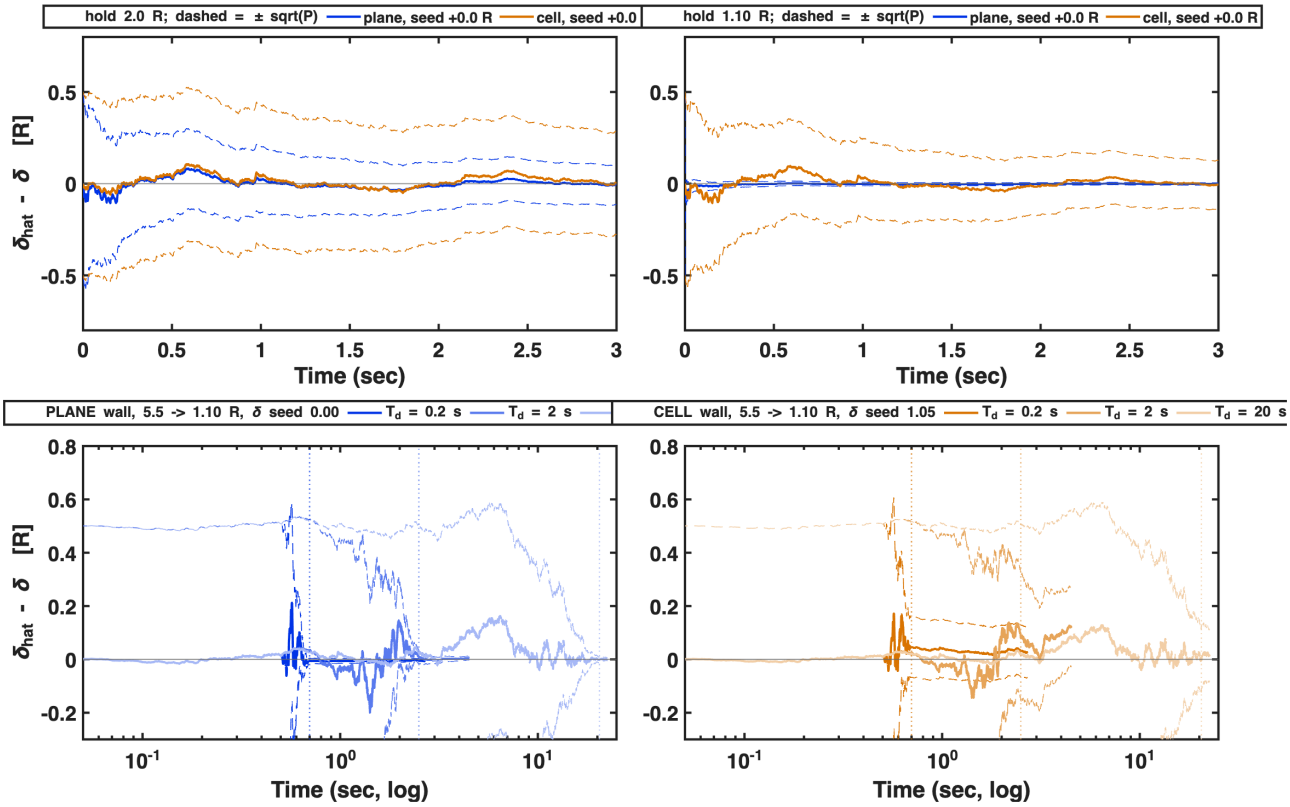

hold 2.0 R; dashed = ± sqrt(P)
plane, seed +0.5 R
cell, seed -0.5 R
hold 1.10 R; dashed = ± sqrt(P)
plane, seed +0.5 R
cell, seed -0.5 R
δhat - δ [R]
Time (sec)
PLANE wall, 5.5 -> 1.10 R, δ seed 0.53
Td = 0.2 s
Td = 2 s
CELL wall, 5.5 -> 1.10 R, δ seed 0.53
Td = 0.2 s
Td = 2 s
Td = 20 s
δhat - δ [R]
Time (sec, log)


hold 2.0 R; dashed = ± sqrt(P)
cell, seed -1.1 R
hold 1.10 R; dashed = ± sqrt(P)
cell, seed -1.1 R
δhat - δ [R]
Time (sec)
PLANE wall, 5.5 -> 1.10 R, δ seed 0.50
Td = 0.2 s
Td = 2 s
CELL wall, 5.5 -> 1.10 R, δ seed 0.00
Td = 0.2 s
Td = 2 s
Td = 20 s
Time (sec, log)


PLANE wall, δ seed 0.00 (= plant) a_true / a_nom a_hat / a_nom (mean ± σ_seed)
CELL wall, δ seed 1.05 (= plant) a_true / a_nom a_hat / a_nom (mean ± σ_seed)
a / a_nom
e_a = a_true - a_hat (mean ± σ_seed)
e_a [a / a_nom]
δ_hat (mean ± σ_seed) ± sqrt(P) δ (plant)
δ [R]
Time (sec)


PLANE, hold 22.2 R, seed 0.00
a_true / a_nom
PLANE, hold 2.0 R, seed 0.00
a_true / a_nom
PLANE, hold 1.10 R, seed 0.00
a_true / a_nom
a_hat / a_nor
a / a_nom
e_a = a_true - a_hat (mean ± σ_seed)
e_a [a / a_nom]
δ_hat (mean ± σ_seed)
± sqrt(P)
δ (plan
δ [R]
Time (sec)


CELL, hold 22.2 R, seed 1.05
a_true / a_nom
CELL, hold 2.0 R, seed 1.05
a_true / a_nom
CELL, hold 1.10 R, seed 1.05
a_true / a_nom
a_hat / a_nom
a / a_nom
e_a = a_true - a_hat (mean ± σ_seed)
e_a [a / a_nom]
δ_hat (mean ± σ_seed)
± sqrt(P)
δ (plant)
δ [R]
Time (sec)


hold 2.0 R; dashed = ± sqrt(P)
plane, seed +0.0 R
cell, seed +0.0
hold 1.10 R; dashed = ± sqrt(P)
plane, seed +0.0 R
cell, seed +0.0 R
δhat - δ [R]
Time (sec)
PLANE wall, 5.5 -> 1.10 R, δ seed 0.00
Td = 0.2 s
Td = 2 s
CELL wall, 5.5 -> 1.10 R, δ seed 1.05
Td = 0.2 s
Td = 2 s
Td = 20 s
Time (sec, log)